

results more disappointing. The typical teenager now spends one-third of his life—not just half his waking hours, but one-third of a day—eight hours—on some sort of electronic device. Does it make him smarter? Richer? More civilized? More coherent? Not that we’ve been able to detect!

Not every technological advance results in an increase in standards of living. Take Twitter, for example. Or nuclear weapons. Or dozens of other innovations and inventions. The Internet, like TV before it, is a great entertainment device. It is also very useful and extremely successful commercially. But has it boosted GDP growth or improved living standards? Probably not. From a GDP growth point of view, the Internet appears to be a dud.

Great advances in living standards and related increases in the value of equities have been driven by big increases in energy use. The Internet did nothing to change that. It was not a “game changer.” The game is the same as it has been since the steam engine was first developed. And now, the “emerging markets” are using oil, too—lots of it. And they’re registering growth rates above 5 percent.

Meanwhile, growth rates in the developed world have declined. In real terms, as mentioned above, U.S. growth in the twenty-first century seems to have fallen back to medieval levels. Why? Who knows? The concept we are presenting here—that developed economies have reached the point of diminishing returns on additional energy inputs—is a maverick idea, at best. Perhaps it is true. Perhaps it isn’t.

A more conventional explanation is that the developed economies are merely in a period of debt consolidation. After 60 years of credit expansion, it is time to reduce debt. That alone could be responsible for the failure of growth and material progress. But why was there so much debt?

First, after the first oil shock in the early 1970s, the economy failed to produce much real growth per capita. Only by going into debt—effectively, spending money they didn’t have—could households increase their standards of living. But we have to remember that the debt itself—and the subsequent deleveraging—are consequences of the same extraordinary energy-driven boom we have been describing. High growth rates over a long period of time convinced almost everyone that standards of living would always go up over time. This allowed both private and public